University of Engineering & Technology, Taxila



Data Warehousing & Mining (LAB)

CEP Report

Name:	**Ahamed Ayyash**
Roll No:	**22-CP-88**
Semester:	**7th**
Section:	**Omega**
Course Instructor:	**Dr. Waqar Ahmad**
Lab Instructor:	**Engr. Usman Rauf**
Dated:	**14/01/2026**

Department Of Computer Engineering

Telecom Customer Churn Analysis

a. Introduction

- **Context:** Telecom churn is a critical metric for customer retention and profitability
- **Churn Definition:** Customers leaving one telecom provider for another
- **Business Impact:** High churn → revenue loss; targeting high-risk customers → cost savings
- **Literature Review:** Refer to previous studies on churn prediction in telecom using ML, DW, and OLAP

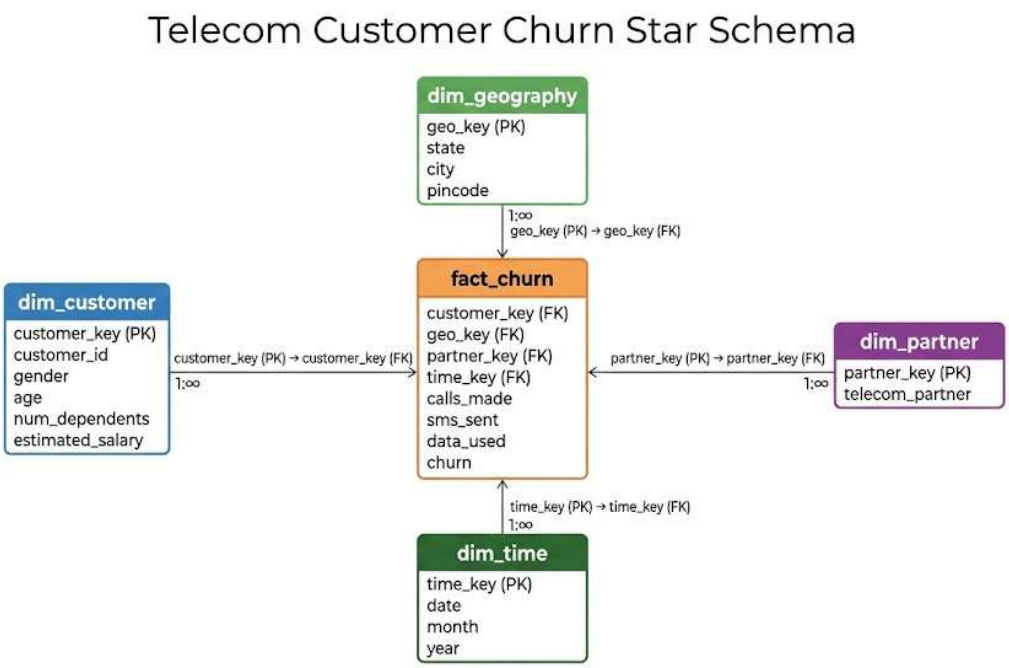
b. DW Architecture

Components:

1. **Staging Layer:** Temporary storage for raw customer data
2. **ETL Layer:** Data cleaning, transformation, surrogate key generation
3. **Warehouse Storage:** Star schema tables (fact_churn, dim_*)
4. **OLAP Engine:** Supports roll-up, drill-down, slice, dice
5. **BI Layer:** Dashboard in Power BI

c. Dimensional Modeling

Star Schema Diagram:



Snowflake Schema:

- dim_geography → normalized into dim_state + dim_city
- dim_partner → normalized into dim_company + dim_plan

Fact Tables:

- fact_churn → contains churn events, usage, and billing metrics

Dimension Tables:

- dim_customer, dim_geography, dim_partner, dim_time

Metadata Design:

- Surrogate keys: customer_key, geo_key, partner_key, time_key
- Load date, source system, ETL batch ID

d. ETL Implementation

```
-- Create dim_customer
CREATE TABLE dim_customer AS
SELECT DISTINCT customer_id,
       gender,
       age,
       num_dependents,
       estimated_salary
FROM stg_telecom_churn;

-- Create dim_geography
CREATE TABLE dim_geography AS
SELECT DISTINCT state, city, pincode
FROM stg_telecom_churn;

-- Add surrogate keys
ALTER TABLE dim_customer ADD COLUMN customer_key SERIAL PRIMARY KEY;
ALTER TABLE dim_geography ADD COLUMN geo_key SERIAL PRIMARY KEY;

-- Create fact_churn
CREATE TABLE fact_churn AS
SELECT c.customer_key,
       g.geo_key,
       p.partner_key,
       t.time_key,
       s.calls_made,
       s.sms_sent,
       s.data_used,
       s.churn
FROM stg_telecom_churn s
JOIN dim_customer c ON s.customer_id = c.customer_id
JOIN dim_geography g ON s.state = g.state AND s.city = g.city AND s.pincode = g.pincode
JOIN dim_partner p ON s.telecom_partner = p.telecom_partner
JOIN dim_time t ON s.date_of_registration = t.date_of_registration;
```

ETL Pipeline:

- Extract → Transform → Load into dimensions
- Populate surrogate keys
- Load fact table with FK references

‘

e. OLAP Analysis

Queries:

- **Roll-up:** Region → State → City

```
SELECT g.state, COUNT(*) AS churn_count
FROM fact_churn f
JOIN dim_geography g ON f.geo_key = g.geo_key
WHERE f.churn = 1
GROUP BY g.state
ORDER BY churn_count DESC;
```

	state text	churn_count bigint
1	Andhra Pradesh	2200
2	Arunachal Pradesh	2236
3	Assam	2217
4	Bihar	2163
5	Chhattisgarh	2122
6	Goa	2213
7	Gujarat	2172
8	Haryana	2222
9	Himachal Pradesh	2246
10	Jharkhand	2333
11	Karnataka	2346
12	Kerala	2194
13	Madhya Pradesh	2278
14	Maharashtra	2216
15	Manipur	2237
16	Meghalaya	2173
17	Mizoram	2266
18	Nagaland	2187
19	Odisha	2257
20	Punjab	2209
21	Rajasthan	2236

- **Drill-down:** Year → Month → Day

```
SELECT t.year, t.month, t.day, COUNT(*) AS churn_count
FROM fact_churn f
JOIN dim_time t ON f.time_key = t.time_key
WHERE f.churn = 1
GROUP BY t.year, t.month, t.day
ORDER BY t.year, t.month, t.day;’
```

Data Output | Messages | Notifications

	year numeric	month numeric	day numeric	churn_count bigint
1	2020	1	1	200
2	2020	1	2	200
3	2020	1	3	200
4	2020	1	4	200
5	2020	1	5	199
6	2020	1	6	200
7	2020	1	7	200
8	2020	1	8	200
9	2020	1	9	200
10	2020	1	10	199
11	2020	1	11	200
12	2020	1	12	200
13	2020	1	13	200
14	2020	1	14	200
15	2020	1	15	199
16	2020	1	16	200
17	2020	1	17	200
18	2020	1	18	200
19	2020	1	19	200

Total rows: 1220 | Query complete 00:00:00.323

- **Slice:** churn = 1

```
SELECT *
FROM fact_churn
WHERE churn = 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 100

	customer_id bigint	telecom_partner text	pincode bigint	date_of_registration timestamp without time zone	calls_made bigint	sms_sent bigint	data_used bigint	churn bigint
1	3	Vodafone	423976	2020-01-01 00:00:00	49	24	193	1
2	4	BSNL	522841	2020-01-01 00:00:00	80	25	9377	1
3	8	BSNL	866786	2020-01-01 00:00:00	87	40	2245	1
4	11	Airtel	776250	2020-01-01 00:00:00	7	7	1275	1
5	17	Vodafone	734068	2020-01-01 00:00:00	16	42	8157	1
6	26	Airtel	651699	2020-01-01 00:00:00	28	48	6434	1
7	32	Airtel	832883	2020-01-01 00:00:00	25	29	1479	1
8	38	BSNL	947518	2020-01-01 00:00:00	69	44	528	1
9	41	Reliance Jio	974052	2020-01-01 00:00:00	16	27	1660	1
10	51	BSNL	517218	2020-01-01 00:00:00	16	27	4865	1
11	58	Vodafone	515124	2020-01-01 00:00:00	20	17	9285	1
12	64	Airtel	234208	2020-01-01 00:00:00	30	48	484	1
13	79	Reliance Jio	491303	2020-01-01 00:00:00	39	35	9625	1
14	81	Vodafone	644267	2020-01-01 00:00:00	27	42	8155	1
15	82	Reliance Jio	567807	2020-01-01 00:00:00	66	10	5821	1
16	88	Airtel	100889	2020-01-01 00:00:00	78	4	-149	1
17	100	Airtel	879464	2020-01-01 00:00:00	42	5	3497	1

Total rows: 48827 | Query complete 00:00:00.169

- **Dice:** Partner = “Reliance Jio” AND data_used < 5

```
SELECT *
FROM fact_churn f
JOIN dim_partner p ON f.partner_key = p.partner_key
WHERE f.churn = 1
AND p.telecom_partner = 'Reliance Jio'
AND f.data_used < 5;
```

Data Output | Messages | Notifications

Showing rows: 1 to 338 | Page No

	customer_key integer	geo_key integer	partner_key integer	time_key integer	calls_made bigint	sms_sent bigint	data_used bigint	churn bigint	fact_key integer	telecom_partner text	partner_key integer
1	3911	36142	3	174	6	46	0	1	373	Reliance Jio	3
2	4789	52145	3	165	34	26	0	1	571	Reliance Jio	3
3	7395	64317	3	184	104	5	0	1	683	Reliance Jio	3
4	12857	14667	3	295	62	41	0	1	1410	Reliance Jio	3
5	21560	14257	3	311	8	0	0	1	2464	Reliance Jio	3
6	26069	4063	3	123	64	6	0	1	3059	Reliance Jio	3
7	40495	228	3	309	67	41	0	1	4972	Reliance Jio	3
8	41880	19430	3	289	6	34	2	1	5210	Reliance Jio	3
9	47571	18526	3	242	67	19	0	1	6014	Reliance Jio	3
10	369	3629	3	319	84	15	0	1	6239	Reliance Jio	3
11	471	21029	3	289	9	11	0	1	6278	Reliance Jio	3
12	1463	44201	3	48	96	26	0	1	6610	Reliance Jio	3
13	1690	62506	3	26	42	33	0	1	6664	Reliance Jio	3
14	6737	56911	3	159	91	40	0	1	7448	Reliance Jio	3
15	31426	16579	3	69	48	40	0	1	11054	Reliance Jio	3
16	35376	28426	3	120	27	42	0	1	11622	Reliance Jio	3
17	37681	26150	3	175	36	13	0	1	11891	Reliance Jio	3

Total rows: 338 | Query complete 00:00:00.706

f. Data Mining Models

1. Classification (Predict churn):

- Algorithm: Decision Tree / Logistic Regression
- Features: age, gender, salary, usage metrics
- Evaluation metrics: Accuracy, Precision, Recall, ROC-AUC

Logistic Regression:

```
Accuracy: 0.5004379602003668
Precision: 0.2018451292409313
Recall: 0.5046403712296984
ROC-AUC: 0.5034173941684891
```

Decision Tree:

```
Accuracy: 0.685380341061506
Precision: 0.20047441058079357
Recall: 0.19032346117101132
ROC-AUC: 0.500142479968514
```

2. Clustering (Segment customers):

- Algorithm: k-Means / EM
- Features: usage patterns, demographic info
- Evaluation metrics: Silhouette score, cluster interpretation

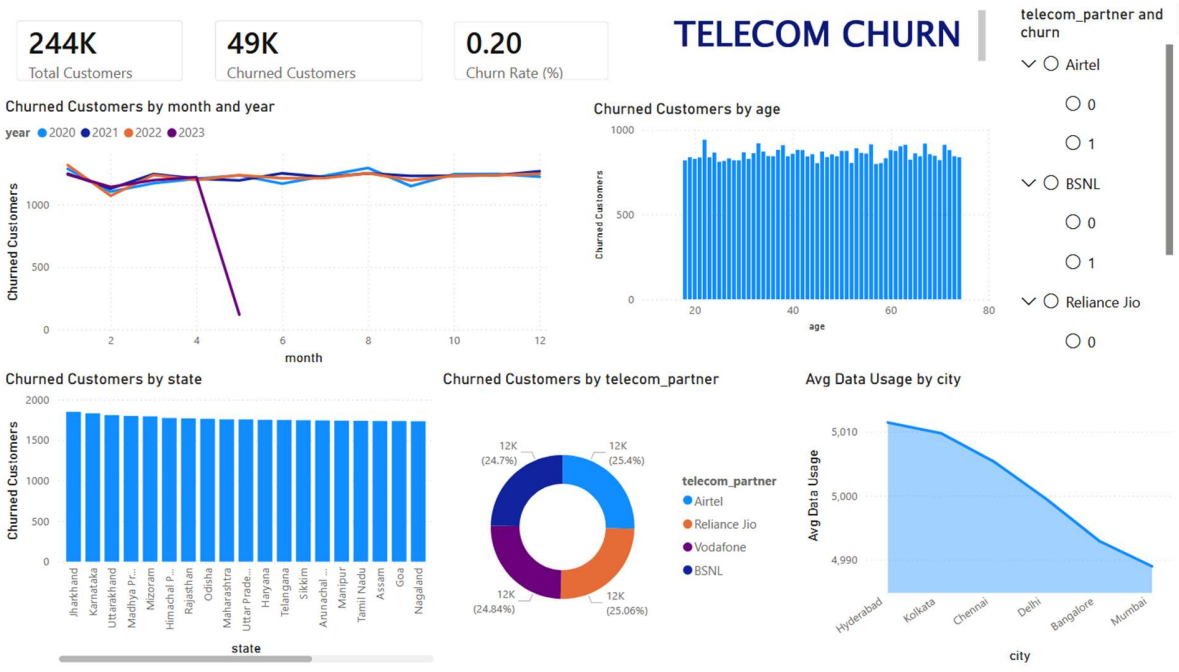
Comparative Analysis: Evaluate which model best predicts high-risk customers

```
Silhouette Score: 0.1087064481133702
```

g. BI Dashboard (Power BI)

Visuals:

1. KPI Cards → Total Customers, Churned Customers, Churn Rate
2. Line Chart → Monthly Churn Trend
3. Bar Chart → Top Churn States
4. Donut Chart → Partner-wise Churn
5. Scatter Plot → Data Usage vs Calls
6. Clustered Column → Age/Gender high-risk segments
7. Slicers → Partner, Churn



Interpretation: Identify high-risk segments, trends, and partners with maximum churn

h. Conclusion

Insights Gained:

- States like Jharkhand, Uttarkhand and karnataka have higher churn
- Low-usage customers are at high risk
- Certain age-gender segments need attention

Limitations:

- Dataset may not capture all churn drivers (e.g., complaints, competitor offers)
- Synthetic data may differ from real telecom data

Recommendations:

- Implement targeted retention campaigns for high-risk segments
- Improve data usage incentives
- Monitor partner-specific churn patterns